

# TrajectoryModel Architecture Description

## 1 Overview

The TrajectoryModel is a neural network designed for predicting vehicle trajectories based on two input images (current and previous frames). It combines an early fusion encoder with an attentive GRU-based decoder, optionally including an auxiliary dynamics prediction head. The encoder includes vertical positional encoding to provide spatial priors for attention. The model processes input images of shape  $(B, 3, 270, 480)$ , where  $B$  is the batch size, and outputs predicted trajectories of shape  $(B, 6, 2)$  for 6 future time steps, along with attention maps and optional auxiliary dynamics predictions.

## 2 Model Components

### 2.1 Utility Blocks

- **ConvGNAct:** A convolutional block consisting of a 2D convolution, group normalization, and ReLU activation.
  - Parameters: input channels, output channels, kernel size (default 3), stride (default 1), padding (default 1), group normalization groups (default 32).
  - Forward: Applies convolution, group normalization, and ReLU activation sequentially.
- **Residual:** A residual block with two ConvGNAct layers.
  - Parameters: Number of channels.
  - Forward: Adds the input to the output of two ConvGNAct layers.

### 2.2 EarlyFusionEncoder

The encoder fuses two input RGB images (current and previous, each of shape  $(B, 3, H, W)$ ) into a single feature map.

- **Input:** Concatenated images of shape  $(B, 6, H, W)$ .
- **Architecture:**
  - *Stem:* ConvGNAct (6 to 32 channels, kernel=7, stride=2), Residual (32), ConvGNAct (32 to 64, stride=2).
  - *Layer1:* Residual (64), ConvGNAct (64 to 128, stride=2).

- *Layer2*: Residual (128), ConvGNAAct (128 to feat\_dim, stride=2).
- *Projection*: 1x1 convolution (feat\_dim to feat\_dim).
- *Vertical Positional Encoding*: Adds a 1-channel positional encoding based on vertical position, with values from -1 to 1 in the useful vertical range (35% to 84% from top).
- **Output**: Feature map of shape (B, feat\_dim + 1, H/16, W/16).

## 2.3 SpatialAttentionDecoder

This module computes spatial attention over the encoder's feature map, conditioned on the decoder's hidden state.

- **Parameters**: Feature dimension (feat\_dim + 1), hidden dimension (hidden\_dim).
- **Components**:
  - Key and value convolutions: 1x1 convolutions on the feature map.
  - Query fully connected layer: Maps hidden state to feat\_dim.
- **Forward**:
  - Computes keys and values from the feature map, reshapes to (B, HW, feat\_dim).
  - Generates query from hidden state, computes attention logits with scaling.
  - Applies softmax to obtain attention weights, computes context vector.
- **Output**: Context vector (B, feat\_dim) and attention map (B, 1, H, W).

## 2.4 AttentiveGRUDecoder

An autoregressive decoder using a GRUCell to predict future trajectory points.

- **Parameters**: Feature dimension (feat\_dim + 1), hidden dimension (hidden\_dim), prediction steps (pred\_steps=6).
- **Components**:
  - Initial hidden and cell state mappings from pooled features.
  - GRUCell with input size (2 + feat\_dim) and hidden size (hidden\_dim).
  - Output head: Linear layers to predict delta x, y for each step.
- **Forward**:
  - Initializes hidden state from pooled feature map.
  - Iteratively applies attention, GRUCell, and output head for each time step.
  - Supports teacher forcing with ground truth trajectories.
- **Output**: Predicted trajectories (B, pred\_steps, 2) and attention maps.

## 2.5 TrajectoryModel

The full model integrates the encoder and decoder, with an optional auxiliary dynamics head.

- **Parameters:** Feature dimension (default 256), hidden dimension (default 256), prediction steps (default 6), use auxiliary dynamics (default False).
- **Components:**
  - EarlyFusionEncoder (outputs  $\text{feat\_dim} + 1$  channels with positional encoding).
  - AttentiveGRUDecoder (uses  $\text{feat\_dim} + 1$ ).
  - Optional auxiliary head: Predicts speed and acceleration from pooled features (uses  $\text{feat\_dim} + 1$ ).
- **Forward:** Processes two input images, returns predictions, auxiliary outputs (if enabled), and attention maps.

## 3 Summary

The TrajectoryModel is a compact yet expressive architecture for trajectory prediction, leveraging early fusion for input processing, vertical positional encoding for spatial priors, spatial attention for feature interaction, and an autoregressive GRU for sequential prediction. Its modular design allows for flexibility in applications requiring vehicle trajectory forecasting.